EV DIAGNOSTIC DECISION SUPPORT SYSTEM

SERVICE MANUAL

Traction Motor / Inverter / HV Thermal System

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This service manual provides diagnostic and repair information for the high-voltage traction system. All procedures must be performed by qualified HV technicians with appropriate PPE. Refer to Engineering Database (EV-DDSS-DB-2026) for component specifications and HV_Power_Distribution_Schematic_v2.png for wiring topology.

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1. Vehicle Overview & System Description

The EV-DDSS demonstrator is a B-segment FWD electric vehicle with a PMSM Traction Motor, liquid-cooled Battery Pack (NMC, 355V nominal), and CAN 2.0B architecture at 500 kbps. The VCM (VCU) coordinates all subsystems including torque demand, thermal management, and fault handling.

Table 1.1: Vehicle Specifications

Parameter	Value	Unit
Vehicle Mass (kerb)	1,450	kg
Gross Vehicle Weight	1,820	kg
Top Speed	140	km/h
Nominal HV Voltage	355	V DC
Battery Capacity	42	kWh
Motor Peak Power	120	kW
Motor Peak Torque	280	Nm
Coolant	G12 EVO 50/50	-
CAN Bit Rate	500	kbps

The ECU network includes Vehicle Controller (VCU), Motor Controller (MCU/Inverter), BMS, and TMM. Diagnostic interface via CAN ID 0x201. Freeze-frame data includes BatteryVoltage, MotorRPM, CoolantTemp.

NOTE: This manual uses multiple naming conventions (e.g., Motor Controller / MCU / Traction Inverter). See Appendix D.

2. High Voltage (HV) Architecture

Nominal 355V DC (250-410V range). Three domains: Battery Pack, Motor Controller (inverter), Traction Motor. Main HV circuit protected by Battery Main Fuse (250A gR) in Battery Pack junction box. Auxiliary Coolant Pump circuit via Auxiliary Fuse F21 (50A, cabin slot 21).

Contactor control: Relay K2 (Main Positive HV) and Relay K5 (Pre-charge/Cooling Pump LV). Pre-charge via K5 resistor, closes K2 when >90% BatteryVoltage reached within 500ms.

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Battery Pack -> Fuse F10 -> Relay K2 -> Connector C21 -> MCU -> Connector C35 -> Traction Motor
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WARNING: HV system up to 410V. Verify de-energized before service. Measure C21 pin1-pin2. Wait 5 min discharge. CAT III meter. Class 00 gloves.

C42 Control Connector (LV) carries CAN, resolver, temp sensors, 12V. See Section 8.

3. Traction Motor / Drive Unit

PMSM-IPM, 120kW/280Nm, resolver (Sin/Cos 2-pole), 3x NTC thermistors in stator. MotorRPM on 0x181 (0.125 rpm/bit, 0-12000). MotorCurrent on 0x182 (0.1A/bit, -400 to +400A).

Table 3.1: Motor Parameters

Parameter	Value	Unit
Type	PMSM-IPM	-
Peak Power	120	kW
Continuous Power	60	kW
Peak Torque	280	Nm
Max RPM	12,000	rpm
Pole Pairs	4 (8 poles)	-
Stator R	0.025	Ohm
Ld/Lq	0.12/0.28	mH
Insulation	Class H (180C)	-
Weight	52	kg

3.1 Motor Winding Test

1. De-energize HV. Disconnect C35.
2. 4-wire DMM: U-V, V-W, W-U. Expected 0.025 Ohm ±10%.
3. Megohmmeter 500V: min 5 MOhm phase-to-ground.
4. Imbalance >5%: replace Traction Motor.
5. C42 resolver check: pins 1-2 and 3-4 = 20-40 Ohm.

4. Motor Controller / Inverter (MCU)

PWM IGBT inverter, 1200V/600A modules, 10kHz switching. FOC with MTPA. Receives torque demand via 0x201 (bytes 0-1, 0.1Nm/bit). Monitors BatteryVoltage (0x182), MotorRPM (0x181), CoolantTemp (0x182), MotorCurrent (0x182).

Table 4.1: MCU Specs

Parameter	Value	Unit
Input Voltage	250-410	V DC
Continuous Current	320	A RMS
Peak Current	400	A RMS
Sw Freq	10 (5-12 var)	kHz
IGBT	1200V/600A	-
Efficiency	97	%
DC Link Cap	1200	uF
Weight	8.5	kg

WARNING: DC link capacitors retain up to 410V for 5 min. Verify C21 pins 1-2 < 5V before opening.

4.1 MCU Power Supply Check

1. Check F18 (< 0.5 Ohm).
2. Close K2 (listen for click, coil ~25 Ohm).
3. Measure C21 pins 1-2: 320-390V.
4. Check 12V at C42 pin 5 (11.5-13.5V).
5. CAN at C42 pins 7-8. Monitor 0x181/0x182.

5. Cooling System / Thermal Management

Water-glycol (G12 EVO, 6.5L), electric Electric Pump (18L/min). Radiator fan. F21 Coolant Pump (50A), Coolant Relay. CoolantTemp on 0x182 (byte 4, scale 1, offset -40C, range -40 to +215C). Normal: 50-85C. Warning: >90C. Shutdown: >105C.

5.1 Coolant Level Check

1. Level surface. HV de-energized.
2. Expansion tank: between MIN-MAX.
3. Color: purple (G12 EVO). Brown = contaminated.
4. Glycol 48-52% via refractometer.

5.2 Pump Function Test

1. Verify F21.
2. Activate K5 via 0x201 byte 3=0x01.
3. Listen for pump. Flow ~18L/min.
4. Monitor CoolantTemp on 0x182.
5. Pump motor resistance: 0.8-1.2 Ohm.

6. Battery Pack / HV Accumulator

NMC 622, 96s1p (12 modules x 8 cells), 355V nom, 42kWh total/37.8kWh usable (10-95% SOC). BMS monitors cell V (±2mV), pack I (shunt, ±0.5%), temp (6 NTC/module). BatteryVoltage on 0x182 (bytes 2-3, bit 16, scale 0.1V/bit).

6.1 HV Isolation Test

1. De-energize. Open service disconnect. Wait 5 min.
2. Megohmmeter 500V: HV+ to chassis, HV- to chassis.
3. Min 500 Ohm/V = 177.5 kOhm at 355V.
4. If < 100 kOhm: inspect for coolant leakage.

7. CAN Communication & Signal Overview

500 kbps CAN 2.0B, single backbone. 120 Ohm termination at each end (VCU, MCU). Stubs < 0.3m.

Table 7.1: CAN Messages

ID	Msg	DLC	Sender	Receiver	Hz
0x181	Motor Status	8	MCU	VCU	10
0x182	HV Status	8	BMS	VCU, MCU	10
0x201	VCU Command	8	VCU	MCU, BMS, TMM	10

Table 7.2: CAN Signals (Intel/little-endian)

Signal	ID	Byte	Start Bit	Len	Scale	Offset	Unit
MotorRPM	0x181	0-1	0	16	0.125	0	rpm
MotorCurrent	0x182	0-1	0	16	0.1	-400	A
BatteryVoltage	0x182	2-3	16	16	0.1	0	V
CoolantTemp	0x182	4	32	8	1	-40	deg C

7.1 CAN Diagnostics

1. Term resistor: C42 pins 7-8 = 60 Ohm ±5%.
2. CAN-H ~2.5V, CAN-L ~2.5V (idle).
3. Traffic: 0x181, 0x182, 0x201 at 10 Hz.

8. Connector Identification & Pinout

Conn	Component	Location	Type	Pins
C21	MCU HV Input	MCU top face	HV orange	2 (HV+,HV-)
C35	Motor Phase Output	MCU bottom face	HV orange	3 (U,V,W)
C42	MCU LV Signal	MCU side face	LV grey 12-way	12

Table 8.2: C21 Pinout

Pin	Signal	Wire Color	Size	Target
1	HV+	Orange	50mm^2	K2/F18
2	HV-	Orange/Black	50mm^2	Batt Neg

Table 8.3: C35 Pinout

Pin	Signal	Color	Size
1	Phase U	Brown	35mm^2
2	Phase V	Black	35mm^2
3	Phase W	Grey	35mm^2

Table 8.4: C42 Pinout

Pin	Signal	Color	Note
1	Resolver SIN+	Green	Rotor position SIN
2	Resolver SIN-	Grn/Wht	-
3	Resolver COS+	Blue	Rotor position COS
4	Resolver COS-	Blu/Wht	-
5	+12V Supply	Red	From VCU 12V/2A
6	GND	Black	Chassis
7	CAN-H	Yellow	CAN 2.0B high
8	CAN-L	Yel/Blk	CAN 2.0B low
9	NTC1 (Stator 1)	White	Motor winding
10	NTC2 (Stator 2)	Wht/Blk	Motor winding
11	CoolantTemp In	Violet	MCU inlet
12	CoolantTemp Out	Vlt/Wht	MCU outlet

9. Fuse & Relay Layout

ID	Rating	Type	Circuit	Location
F18	250A	gR (semiconductor)	MCU HV Main	Battery Pack J-Box
F21	50A	Medium-blow LV	Cooling Pump	Cabin Fuse Panel Slot 21

ID	Function	Type	Location	Coil R
K2	Main Positive HV	SPST NO 450V/400A	Battery J-Box	~25 Ohm
K5	Cooling Pump LV	SPST NO 12V/30A	Cabin Panel Slot 22	~85 Ohm

10. Diagnostic Error Code Reference

10.1 P0A94 - MCU HV Input Fault

Severity: HIGH

Description: Open K2, damaged C21, blown F18, missing MotorRPM on 0x181. Cross-ref: S11.1, S8, S9, S7.

10.2 P1C21 - Motor Overtemperature

Severity: MEDIUM

Stator >150C via NTC1 (C42 pin 9). Check CoolantTemp on 0x182, F21, K5, coolant level. Cross-ref: S11.2, S5.2, S9.

10.3 P2B07 - CAN Lost (MCU->VCU)

Severity: HIGH

Check C42 pins 7-8. 60 Ohm term. 0x181/0x182/0x201 traffic. Cross-ref: S11.3, S7.1, S8.

10.4 P3F10 - Cooling System Degraded

Severity: MEDIUM

CoolantTemp >90C. F21, K5, pump, coolant level. >105C = shutdown + P1C21. Cross-ref: S11.4, S5, S9.

11. Troubleshooting Procedures

11.1 P0A94 - Motor Does Not Rotate

1. BatteryVoltage on 0x182: expected 320-390V.
2. F18 continuity: < 0.5 Ohm. DMM.
3. K2: listen for click. Coil ~25 Ohm.
4. C21 pins 1-2: 320-390V. CAT III DMM.
5. MotorRPM on 0x181: 0 when stationary.
6. C42 pins 1-2 resolver: 20-40 Ohm.
7. MCU fault log via 0x201 diag service.

Table 11.1: P0A94 Measurements

Test Point	Tool	Expected	If Failed
F18	DMM	< 0.5 Ohm	Replace F18 250A
K2 coil	DMM	~25 Ohm	Replace K2
C21 pins 1-2	CAT III DMM	320-390V	Check wiring W101
C42 pins 7-8	DMM	60 Ohm	Check CAN term
0x181 MotorRPM	CAN tool	0 rpm	MCU/resolver

11.2 P1C21 - Overtemperature

1. CoolantTemp on 0x182: 50-85C.
2. Coolant level: MIN-MAX.
3. F21: < 0.5 Ohm. K5: ~85 Ohm.
4. Activate pump: 0x201 byte 3=0x01.
5. Pump resistance: 0.8-1.2 Ohm.
6. NTC at C42 pin 11: ~10kOhm @25C.

11.3 P2B07 - CAN Failure

1. 12V at C42 pin 5: 11.5-13.5V.
2. C42 pins 7-8: 60 Ohm.
3. CAN-H ~2.5V, CAN-L ~2.5V.
4. Traffic: 0x181, 0x182, 0x201 @10Hz.
5. Inspect C42 pins for damage.

11.4 P3F10 - Cooling Degraded

1. CoolantTemp on 0x182: >90C.
2. F21 continuity.
3. Activate K5 via 0x201.
4. Coolant level + glycol 48-52%.
5. Radiator fan at 85C.

12. Preventive Maintenance

Interval	Item	Ref
10k/12mo	Coolant level, condition	S5.1
20k/24mo	HV isolation test	S6.1
30k/36mo	Coolant replacement	S12.2
40k/48mo	Motor winding test	S3.1
50k/60mo	CAN integrity test	S7.1
Annual	Fuse/relay (F18,F21,K2,K5)	S9
Annual	Connector (C21,C35,C42)	S12.1

12.1 Connector Inspection

- C21: check overheating. Torque 12Nm (M8).
- C35: check corrosion. Torque 8Nm (M6).
- C42: check bent pins. Dielectric grease.
- Wiring: W101, W102, W201 for chafing.

12.2 Coolant Replacement

1. HV de-energized. Coolant < 40C.
2. Drain. Fill G12 EVO 50/50.
3. Run pump via 0x201 byte 3=0x01, 5 min.
4. Bleed air. Verify 48-52% glycol.

Appendix A: Schematic Reference

See HV_Power_Distribution_Schematic_v2.png (DWG: EV-DDSS-SCH-2026-002, Rev B) for wiring topology including pin numbers, wire colors, wire IDs (W101-W208), ground points, and CAN routing.

Appendix B: Torque Specifications

Fastener	Torque	Tool	Notes
C35 bolts (M6)	8 Nm	Torque wrench	Loctite 243
C21 bolts (M8)	12 Nm	Torque wrench	HV critical
MCU mount (M10)	25 Nm	Torque wrench	3 stages
Batt terminals (M8)	18 Nm	Torque wrench	HV critical
Pump mount (M6)	10 Nm	Torque wrench	Isolation mounts
Temp sensor (M12)	15 Nm	Torque wrench	PTFE compound

Appendix C: Recommended Tools

Tool	Purpose	Spec
CAT III DMM	HV V/R/cont	1000V CAT III
Megohmmeter	Isolation test	500-1000V DC
CANalyzer/CANnect	CAN monitoring	500kbps CAN 2.0B
Oscilloscope	CAN sig analysis	100MHz
HV Gloves Class 00	Shock protection	500V, test <6mo
Refractometer	Glycol %	G12 EVO scale
Torque Wrench	Fasteners	5-50 Nm

Appendix D: Abbreviations

Abbr	Full Term
BMS	Battery Management System
CAN	Controller Area Network
DDSS	Diagnostic Decision Support System
DMM	Digital Multimeter
DTC	Diagnostic Trouble Code
ECU	Electronic Control Unit
FOC	Field-Oriented Control
HV	High Voltage
IGBT	Insulated-Gate Bipolar Transistor
IPM	Interior Permanent Magnet
LV	Low Voltage (12V)
MCU	Motor Controller Unit (Traction Inverter)
NMC	Nickel Manganese Cobalt
NTC	Negative Temperature Coefficient
PMSM	Permanent Magnet Synchronous Motor
PPE	Personal Protective Equipment
TMM	Thermal Management Module
VCU	Vehicle Control Unit